



Management and Cost Accounting – Activity Based Costing

Exercise 4.2

The *Technetium Gear Company* manufactures and distributes two types of gaming chairs: the "Titan Throne" (a premium, ergonomic chair) and the "Arena X" (a basic, lightweight chair). *Technetium Gear* is concerned about the profitability of each chair model under their current costing system. They currently use a traditional costing method with unit output as the allocation base, but suspect it might not capture the true cost drivers for each chair type. Management has instructed the head of cost accounting to perform a comparative analysis using both traditional costing and activity-based costing (ABC) based on the data provided in Tables 1 and 2. The indirect production costs are driven by machine hours, while delivery costs are driven by the number of shipments.

Required

1. Calculate the overhead rate and total cost per unit for Titan Throne and Arena X using the traditional costing approach with unit outputs as the allocation base.
2. Calculate the total cost per unit for Titan Throne and Arena X using the activity-based costing approach.
3. Compare and interpret the results of (1) and (2) in detail. What advice would you give to management?

Table 1

	Titan Throne	Arena X
Output in units	4,800.00	10,200.00
Costs/cost drivers:		
Direct materials	€650,000.00	€860,000.00
Direct labour	€480,000.00	€600,000.00
Total machine hours	3,000	7,800
Number of shipments	40	250

Table 2

Activity pool	Annual cost €	Cost driver	Amount
Production	€450,000.00	Machine hours	10,800.00
Delivery	€200,000.00	Number of shipments	290
Total	€650,000.00		

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Task 1:

Calculate the overhead rate and total cost per unit for Titan Throne and Arena X using the traditional costing approach.

	Titan Throne	Arena X	
Output in units	4,800	10,200	
Costs/cost drivers:			
Direct materials	€ 650,000	€ 860,000	Direct costs
Direct labour	€ 480,000	€ 600,000	
Total machine hours	3,000	7,800	
Number of shipments	40,00	250,00	
Activity pool	Annual cost €	Cost driver	Units
Production	€ 450,000	Machine hours	10,800
Delivery	€ 200,000	Number of shipments	290
Total	€ 650,000		

Step 1: Calculate the overhead rate

Overhead rate: €43.33

Step 2: Assign costs to the products

Total cost/unit

Titan Throne	Arena X
€ 278.75	€ 186.47

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Task 2: Calculate the total cost per unit for Titan Throne and Arena X using the activity-based costing approach.

	Titan Throne	Arena X	
Output in units	4,800	10,200	
Costs/cost drivers:			
Direct materials	€ 650,000	€ 860,000	
Direct labour	€ 480,000	€ 600,000	
Total machine hours	3,000	7,800	
Number of shipments	40,00	250,00	
Activity pool	Annual cost €	Cost driver	Units
Production	€ 450,000	Machine hours	10,800
Delivery	€ 200,000	Number of shipments	290
Total	€ 650,000		

Direct costs

Indirect costs

Step 1: Calculate the cost driver rates

Activity pool	Cost driver rate
Production	€ 41.67
Delivery	€ 689.66

Step 2: Allocate indirect costs to the products

Total overhead

Titan Throne	Arena X
€ 152,586.21	€ 497,413.79

Exercise 4.2

Task 2: Calculate the total cost per unit for Titan Throne and Arena X using the activity-based costing approach.

Step 3: Calculate the total costs of the products

	Titan Throne (ABC)	Arena X (ABC)
Total cost/unit	€ 267.21	€ 191.90

Exercise 4.2

Task 3: Compare and interpret the results of (1) and (2) in detail. What advice would you give to management?

	Titan Throne (TC)	Titan Throne (ABC)	Arena X (TC)	Arena X (ABC)
Total cost/unit	€ 278.75	€ 267.21	€ 186.47	€ 191.90

	Traditional costing	ABC	Difference
Titan Throne	€ 278.75	€ 267.21	€ 11.54
Arena X	€ 186.47	€ 191.90	€ -5.43

Under the traditional costing method, we allocated indirect costs based on output in units. Therefore, we calculated the overhead rate and allocated indirect costs according to the output of units. Under the ABC method, we identify various activities that cause overhead costs and allocate these indirect costs to products based on the consumption of those activities. Under ABC, the cost per unit of Titan Throne decreases while the cost per unit of Arena X increases compared to the traditional method. This suggests that costs under the traditional method were distorted, and that Arena X consumes a larger proportion of resources and, hence, costs compared to the Titan Throne.

- Potential consequences in decision-making: Produce more Titan Thrones and fewer Arena X.
- Re-evaluate pricing strategies
- Etc.

ABC vs. Traditional Costing

- Pros:
 - More accurate costing of activities and, hence, more accurate product costs
 - Improved decision-making
 - Improvement in product and process design
 - Better understanding of overhead costs

- Contras:
 - Costly & timely (implementation, measurements, maintenance, etc.)
 - Difficult to understand
 - Potentially misleading results (if not maintained & updated regularly)
 - Resistance of workers during and after implementation

Major, M. & Hopper, T. (2005). Managers Divided: Implementing ABC in a Portuguese Telecommunications Company. *Management Accounting Research*, 16(2), 205-229.

Exercise 12.12

Pagnol-Carrelages SNC is a small distributor of marble tiles. Pagnol-Carrelages identifies its three major activities and cost pools as ordering, receiving and storage, and shipping, and reports the following details for 2022:

Activity	Cost driver	Quantity of cost driver	Cost per unit of cost driver
1 Placing and paying for orders of marble tiles	Number of orders	500	€50 per order
2 Receiving and storage	Number of loads moved	4000	€30 per load
3 Shipping of marble tiles to retailers	Number of shipments	1500	€40 per shipment

Pagnol-Carrelages buys 250 000 marble tiles at an average cost of €3 per tile and sells them to retailers at an average price of €4 per tile. Fixed costs are €40 000.

Required:

- 1) Calculate Pagnol-Carrelages' operating profit for 2022.
- 2) For 2023, retailers are demanding a 5% discount off the 2022 price. Pagnol-Carrelages' suppliers are only willing to give a 4% discount. Pagnol-Carrelages expects to sell the same quantity of marble tiles in 2023 as it did in 2022. If all other costs and cost driver information remain the same, what will Pagnol-Carrelages' operating profit be in 2023?
- 3) Suppose further that Pagnol-Carrelages decides to make changes in its ordering, and receiving and storing practices. By placing long-term orders with its key suppliers, it expects to reduce the number of orders to 200 and the cost per order to €25. By redesigning the layout of the warehouse and reconfiguring the crates in which the marble tiles are moved, Pagnol-Carrelages expects to reduce the number of loads moved to 3125 and the cost per load moved to €28. Will Pagnol-Carrelages achieve its target operating profit of €0.30 per tile in 2023? Show your calculations.

Exercise 12.12

Task 1:

Calculate Pagnol-Carrelages' operating profit for 2022.

	(1) Total for 250.000 tiles	Per unit = (1)/250.000
Operating profit/loss	€5.000,00	€0,02

Exercise 12.12

Task 2:

For 2023, retailers are demanding a 5% discount off the 2022 price. Pagnol-Carrelages' suppliers are only willing to give a 4% discount. Pagnol-Carrelages expects to sell the same quantity of marble tiles in 2023 as it did in 2022. If all other costs and cost driver information remain the same, what will Pagnol-Carrelages' operating profit be in 2023?

	(1) Total for 250.000 tiles	Per unit = (1)/250.000
Operating profit/loss	-€15.000,00	-€0,06

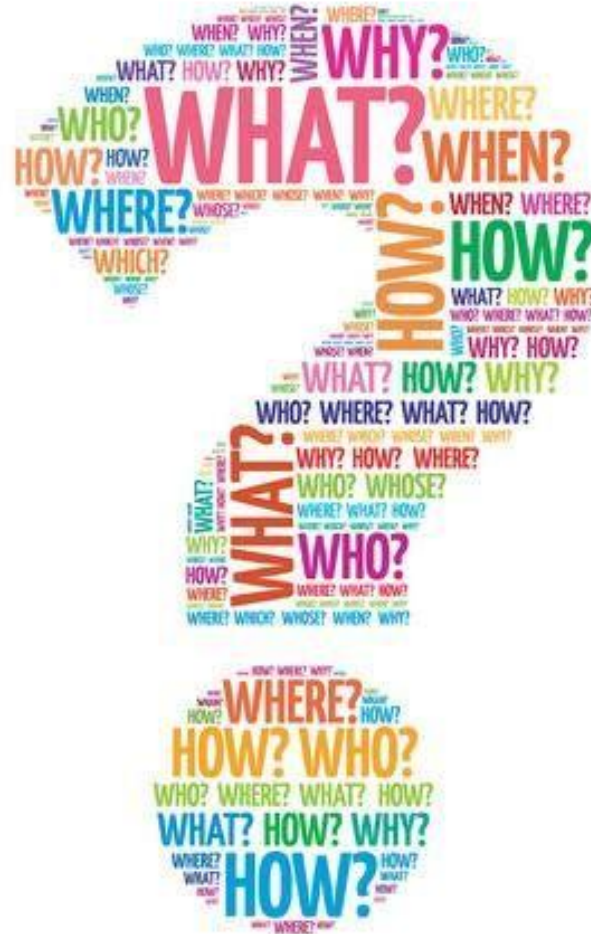
Exercise 12.12

Task 3:

Suppose further that Pagnol-Carrelages decides to make changes in its ordering, and receiving and storing practices. By placing long-term orders with its key suppliers, it expects to reduce the number of orders to 200 and the cost per order to €25. By redesigning the layout of the warehouse and reconfiguring the crates in which the marble tiles are moved, Pagnol-Carrelages expects to reduce the number of loads moved to 3125 and the cost per load moved to €28. Will Pagnol-Carrelages achieve its target operating profit of €0.30 per tile in 2023? Show your calculations.

	(1) Total for 250.000 tiles	Per unit = (1)/250.000
Operating profit/loss	€37.500,00	€0,15

Questions?



<https://www.pinterest.de/pin/799037158920259827/>

References

Bhimani, A., Horngren, C. T., Datar, S. M., & Rajan, M. V. (2023). *Management and Cost Accounting* (8th ed.). Pearson.

Major, M. & Hopper, T. (2005). Managers Divided: Implementing ABC in a Portuguese Telecommunications Company. *Management Accounting Research*, 16(2), 205-229.